



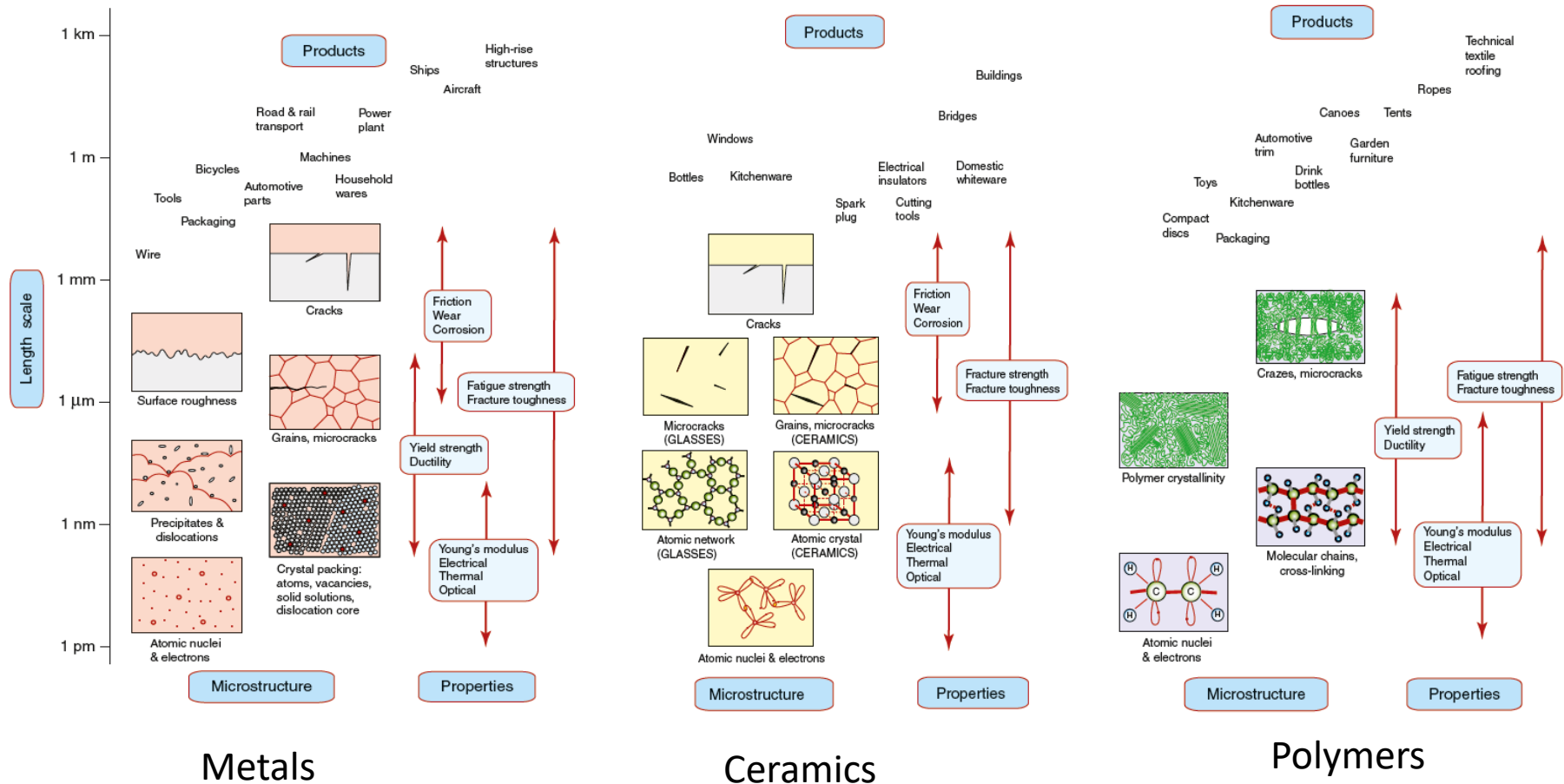
Sustainable Engineering 4: Materials processing

Presented by Dr Adijat 'Wumi Inyang

Microstructure and properties

- Microstructure of the material is determined by the processing route (manufacture method and heat treatment)
- Material performance characteristic depend on the microstructure
- We choose the processing route for the material to achieve both the shape and the microstructure
- Consider the environmental impact of manufacture

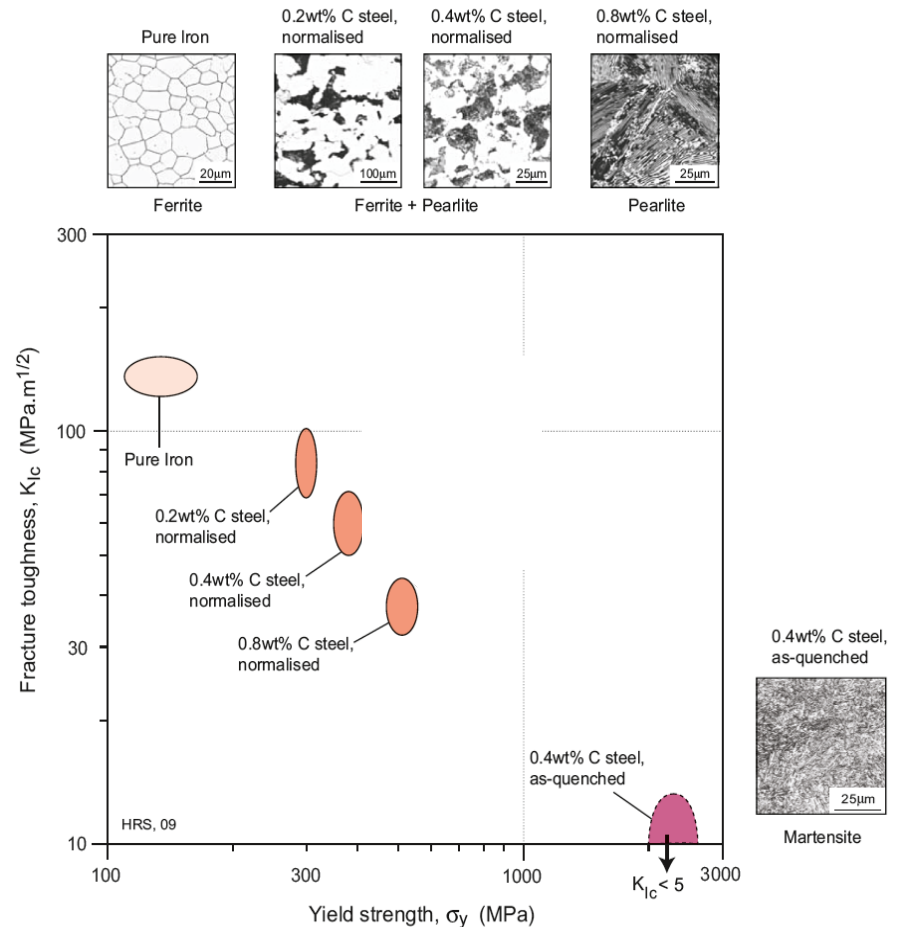
Microstructure and properties



Effect of heat treatment

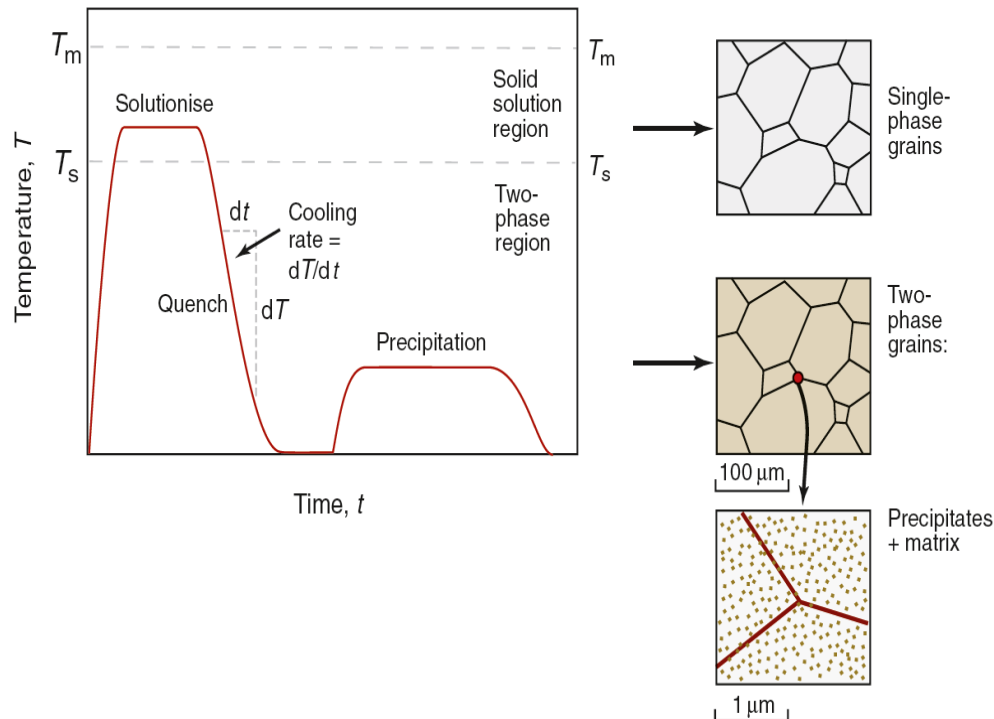
- Normalizing
 - Slow cooling from high temperature producing a microstructure of lower strength but high toughness
 - No follow-up heat treatment

- Quench and temper
 - Cooled faster than the critical cooling rate producing a martensitic microstructure with high strength and low toughness
 - Then reheated at an intermediate temperature to restore toughness



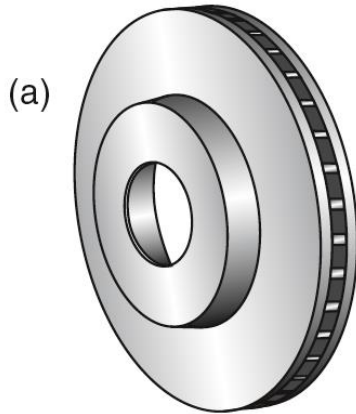
Heat treatment of metals

In a precipitation hardening heat treatment, a material is heated to high temperature, cooled at a controlled rate, and then reheated to an intermediate temperature

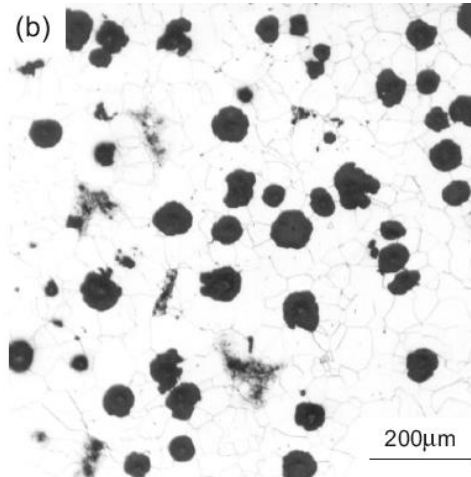
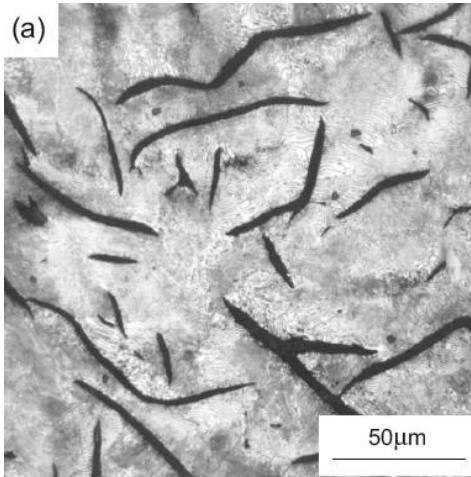


- The figure illustrates the thermal profile and resulting microstructure for a heat treatment process
- A solid solution is formed at high temperature followed by precipitation hardening at an intermediate temperature after being cooled

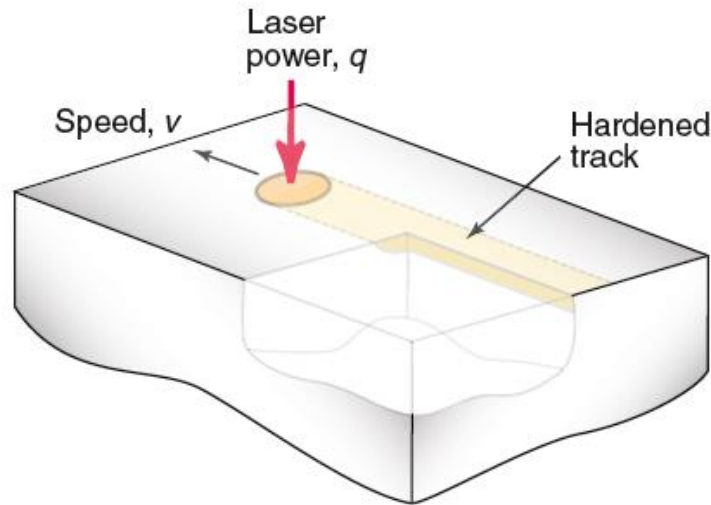
Cast iron disk brake



Most ferrous alloys offer the hardness, wear resistance, toughness, and high $T_{\max}$ required for a disk brake – the composition of cast iron allows for significant microstructural modifications with minor addition of alloying elements. Gray cast iron has excellent machinability, making it a good option for a disk brake



- Poisoning is the addition of a small amount of cerium or magnesium, sometimes with the addition of an annealing treatment, which improves the morphology of graphite in iron
- This gives cast iron greater toughness for a wider range of service conditions

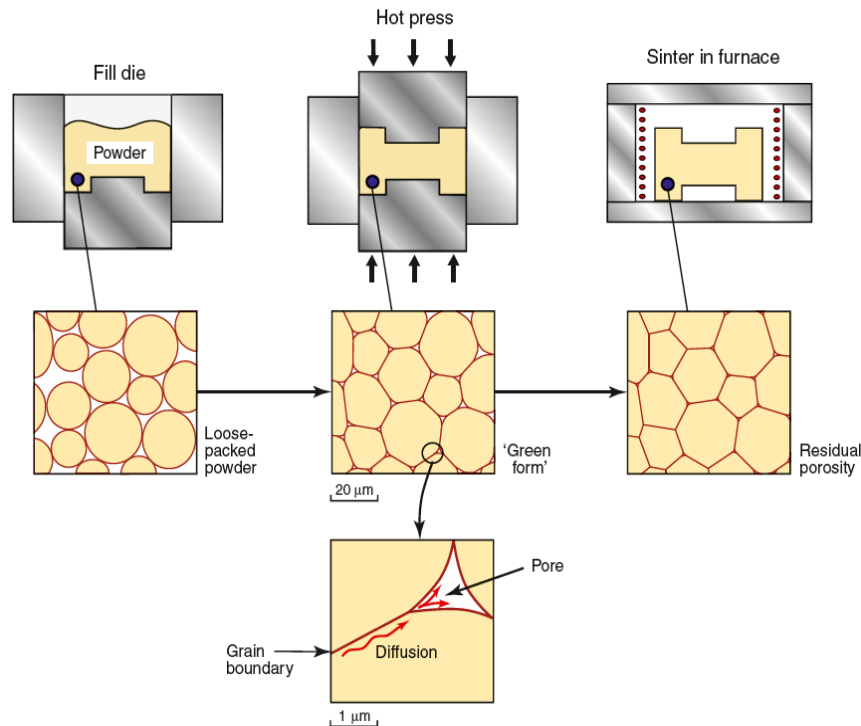


Surface treatments exploit many different mechanisms and processes to change the surface microstructure and properties

Laser hardening is a surface treatment process that modifies microstructure – the traversing laser beam induces a rapid thermal cycle, causing phase changes on both heating and cooling – the track below the path of the laser has a different microstructure with high hardness

Glass and ceramics

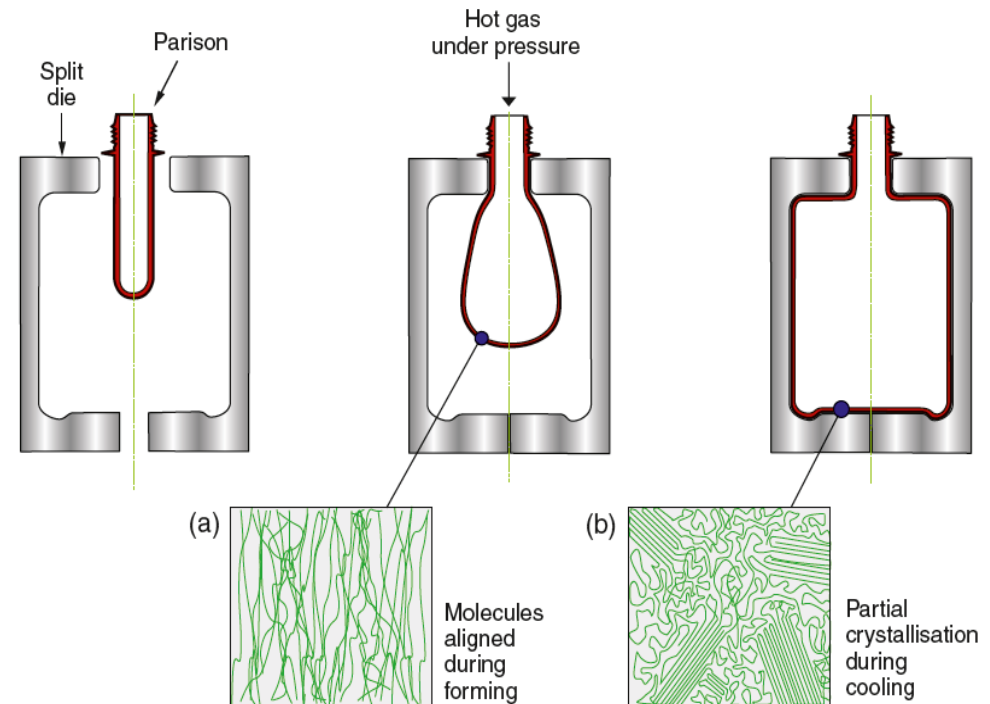
Because of their susceptibility to cracking, defect control is the most important aspect of processing glasses and ceramics – while some thermal processes can increase the toughness of glasses, changes in composition is generally used to alter the properties of ceramics



Ceramics can be shaped by filling a mold with loose powder and compacting it – the main microstructural evolution is the shrinkage of porosity during sintering by diffusion processes

Polymers and elastomers

- Processing can play a significant role in determining the final molecular arrangement and thus properties if the molecules are first aligned mechanically by the flow induced during the shaping process, and cooling is rapid enough to freeze the molecular alignment



- The most dramatic impact of processing on polymer properties is in making fibers
- Drawing fibers creates significant molecular alignment and the covalent bonding along the chain is exploited to markedly increase the specific strength and stiffness of the polymer

Environmental impact – example 1

Manufacturing a PET bottle

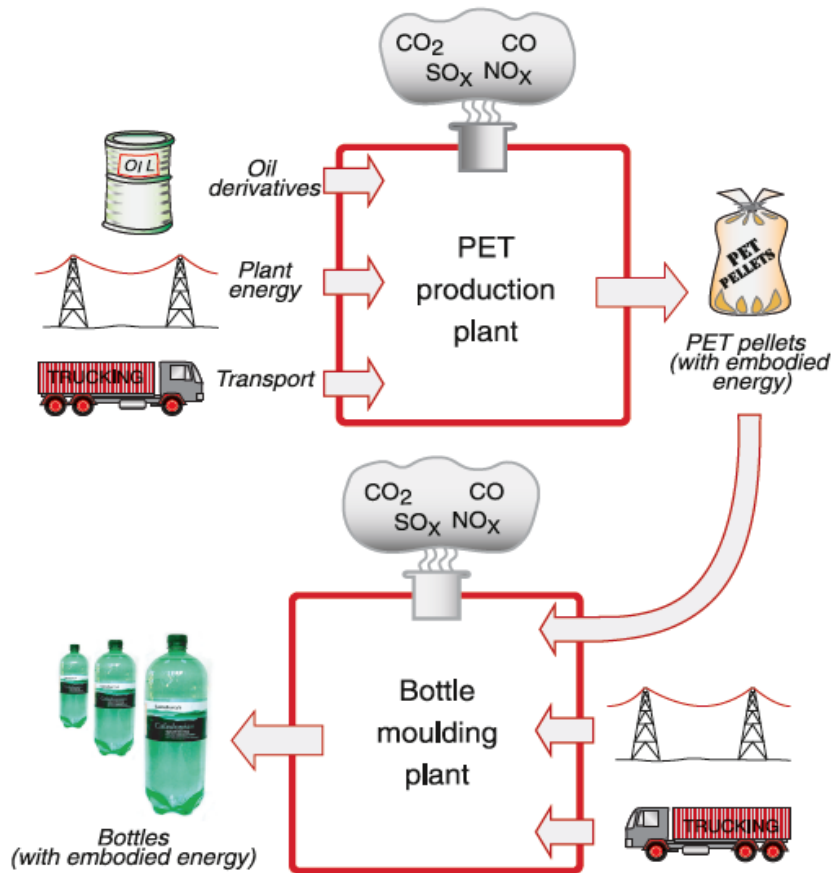


Figure 20.4

- **Stage 1 – material production**

Embodied energy, H_m , of PET = 85 MJ/kg

Carbon footprint = 4kg CO₂ / kg

- **Stage 2 – bottle production**

Processing energy, H_p =

3 MJ/kg for extrusion of parison

3 MJ/kg for blow moulding bottle

(energy efficiency – 33%)

$H_p = (3\text{MJ} + 3\text{MJ})/0.33 = 18\text{MJ/kg}$

Environmental impact – example 1

Manufacturing a PET bottle

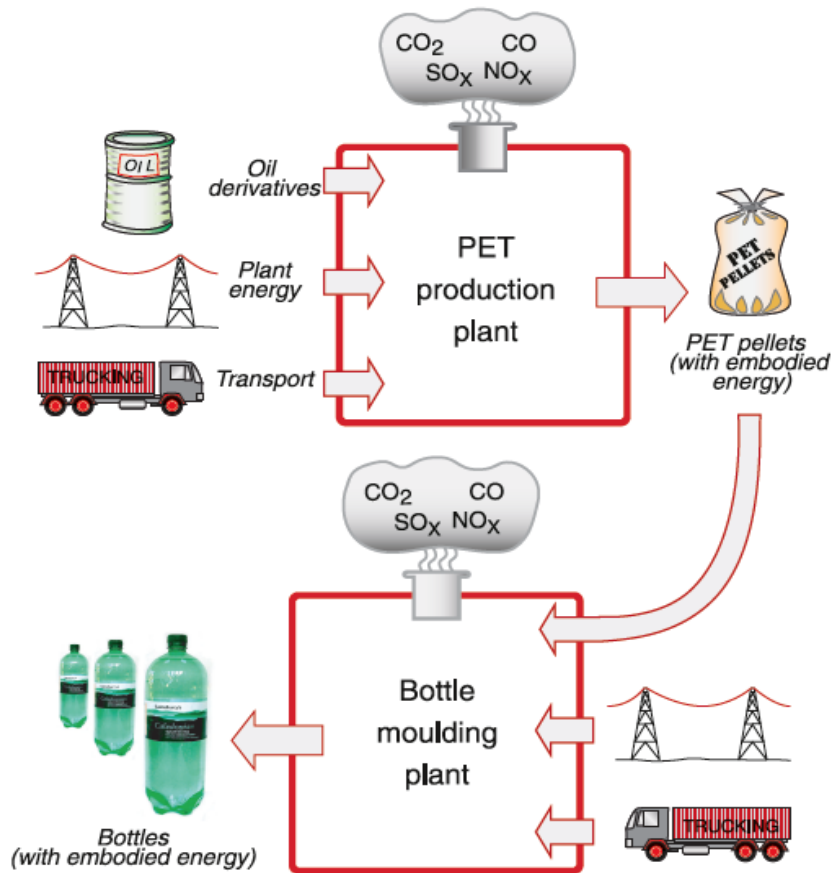


Figure 20.4

- **Total energy input to make 1 bottle**
Mass of 1 bottle = 67g

$$H_m = 85\text{MJ} \times 0.067\text{kg} = \mathbf{5.7\text{MJ}}$$

$$H_p = 18\text{MJ} \times 0.067\text{kg} = \mathbf{1.2\text{MJ}}$$

$$\mathbf{\text{Total energy} = 5.7 + 1.2 = 6.9\text{MJ}}$$

- Energy embedded in the product is dominated by the **material**
- Using recycled PET ($H_m = 55\text{MJ/kg}$) would reduce total energy by 29%

Environmental impact – example 1

Manufacturing a PET bottle

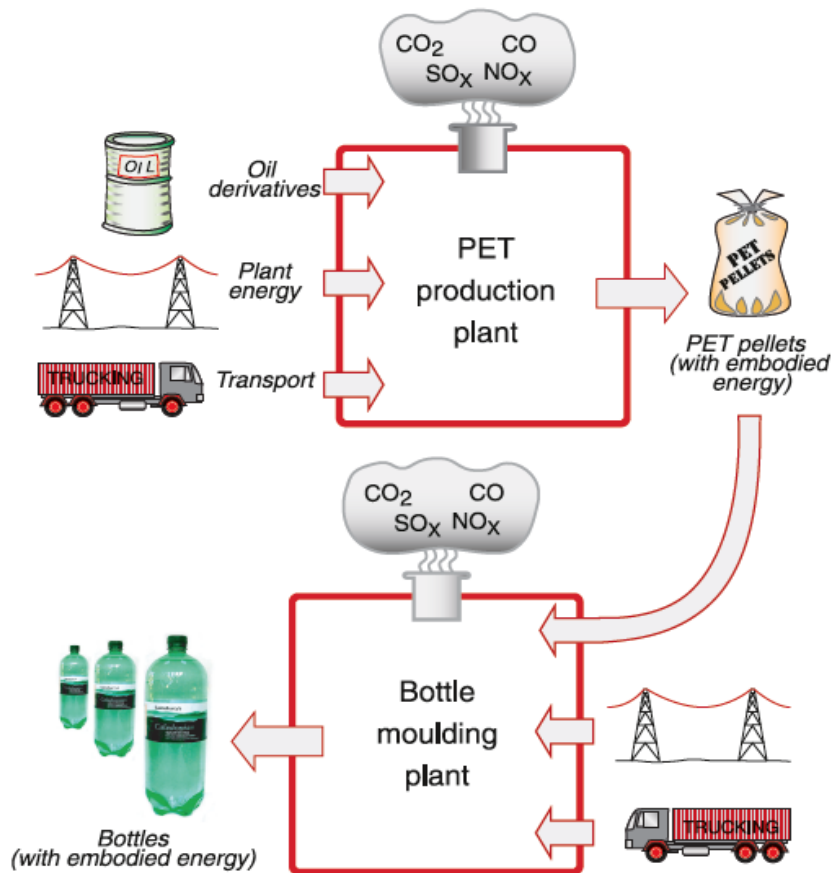


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In this example we have:

Assumed there is very little waste PET

Not considered the transport impact

Not considered the impact of machinery

Environmental impact – example 2

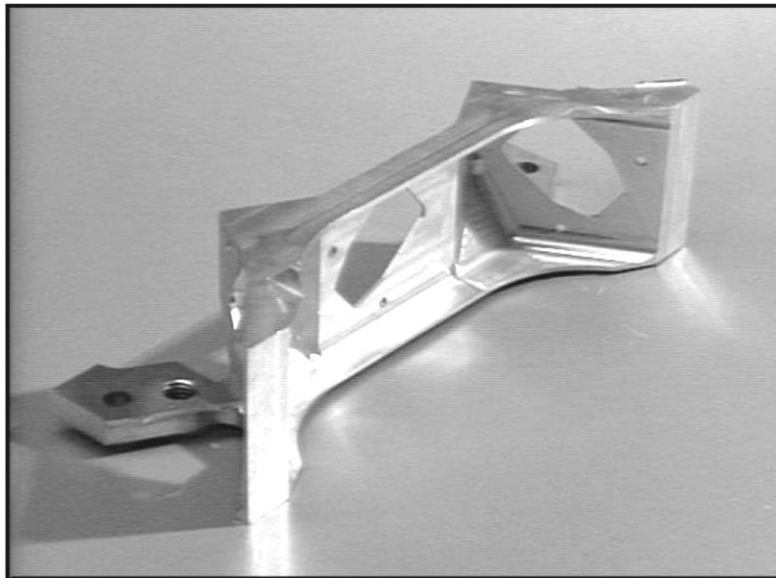
Manufacturing an aluminium bracket

Part mass = 0.444kg

Starting block = 30 x 12 x 9 cm

Density of Al = 2.7 g/cm³

Mass of block = 8.748kg



- **Stage 1 – material production**

Embodied energy, H_m , of Al = **200 MJ/kg**

Carbon footprint = 12kg CO₂ / kg

- **Stage 2 – machining**

Processing energy, H_p =

3 MJ/kg for milling machine

(energy efficiency – 33%)

Material removed = 8.304kg (95%)

(Buy/fly ratio = 20. i.e. only 1/20 used)

$$H_p = 3\text{MJ} / 0.33 = \mathbf{9\text{MJ/kg}}$$

Environmental impact – example 2

Manufacturing an aluminium bracket

Part mass = 0.444kg

Starting block = 30 x 12 x 9 cm

Density of Al = 2.7 g/cm³

Mass of block = 8.748kg

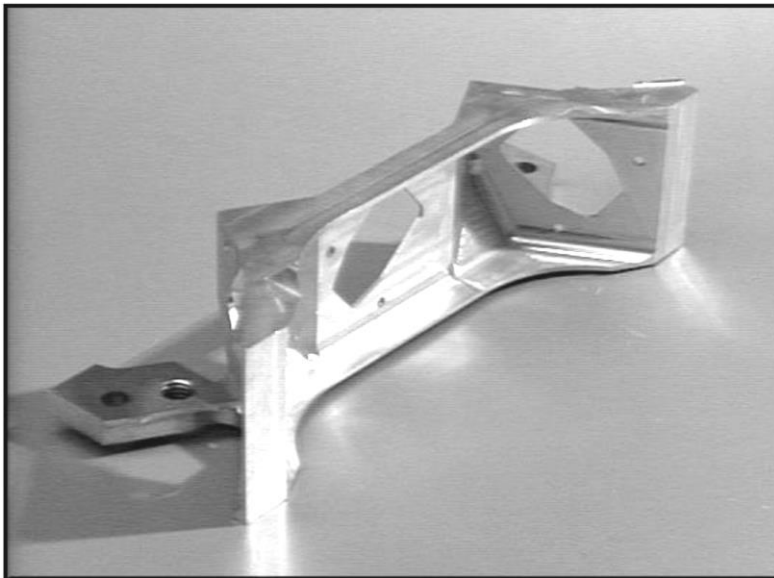
- **Total energy input to make 1 bracket**

$$H_m = 200\text{MJ} \times 8.748\text{kg} = \mathbf{1750\text{MJ}}$$

$$H_p = 9\text{MJ} \times 8.304\text{kg} = \mathbf{75\text{MJ}}$$

$$\mathbf{\text{Total energy} = 1750 + 75 = 1825\text{MJ}}$$

- Energy embedded in the product is dominated by the **material**
- Large amount of **waste** metal = 95%
- 3D printing would not reduce energy but would reduce waste, giving more parts from the starting material



Summary



- Processing routes determine the microstructure of the material – and so the properties
- Choose the processing route to achieve material performance
- Energy required and speed of processing depend on the processing route
- In the examples given here, the energy invested in the product is dominated by the raw material
- Select materials carefully – reuse of material can greatly reduce the environmental impact